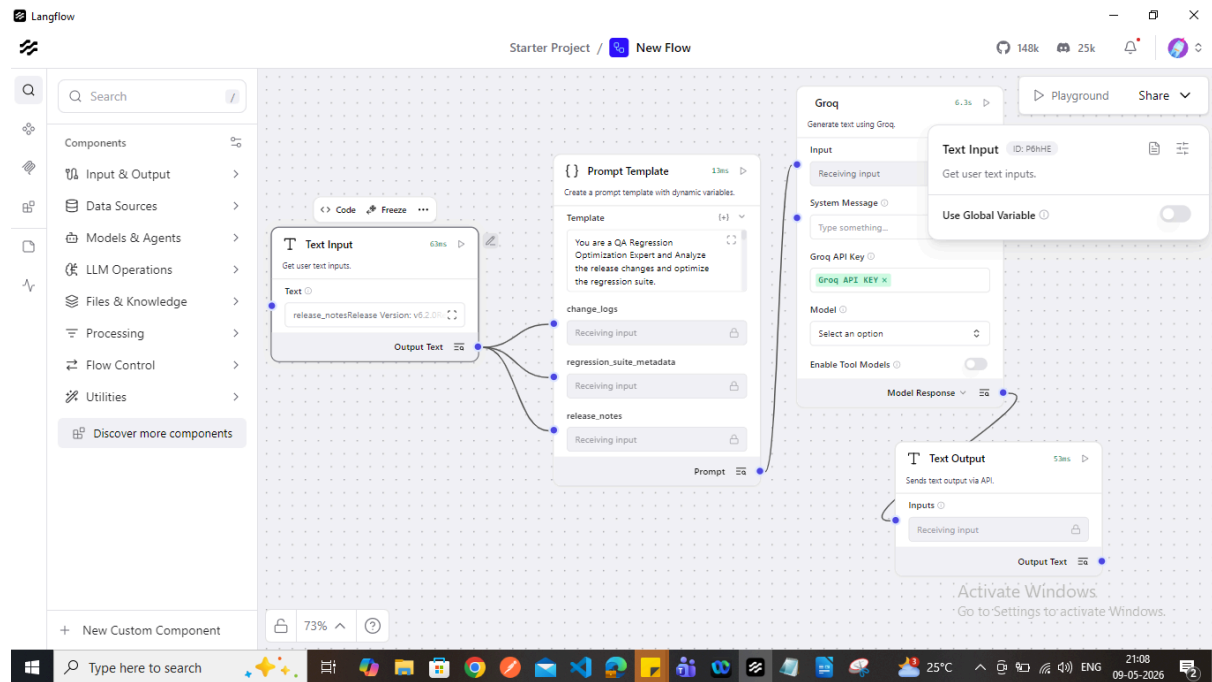


Intelligent Regression Impact Analysis and Test Optimization System using Langflow

Created the Problem Statement into POC using LangFlow



1) Added "Text Input" in Langflow and added the below as input in LangFlow

release_notes

Release Version: v6.2.0
Release Date: 10-May-2026

New Features:

- Multi-Factor Authentication (MFA) added for user login
- Payment retry mechanism enhanced
- Cart page redesigned with responsive UI

Enhancements:

- Session timeout increased from 15 mins to 30 mins
- Improved API response handling in checkout module

Bug Fixes:

- Fixed duplicate payment issue during retry
- Fixed cart item quantity update issue
- Fixed intermittent logout issue

Modules Impacted:

- Login
- Authentication
- Payment

- Cart
- Checkout
- Session Management

change_logs

Change Log Summary:

[LOGIN MODULE]

- Added MFA OTP validation
- Added resend OTP functionality
- Updated login API response structure

[PAYMENT MODULE]

- Modified payment retry logic
- Added retry limit validation
- Updated timeout handling

[CART MODULE]

- Redesigned cart UI
- Updated quantity increment/decrement functionality
- Improved mobile responsiveness

[SESSION MANAGEMENT]

- Updated session timeout configuration
- Added automatic inactivity logout validation

Database Changes:

- Added MFA_OTP table
- Updated PAYMENT_TRANSACTION table

regression_suite_metadata

Regression Suite Details:

Total Scripts: 120

Automation Framework: Selenium + Pytest

Script Metadata:

1. Script Name: test_login_valid.py

Module: Login

Priority: Critical

Last Execution: Passed

Execution Time: 2 mins

2. Script Name: test_login_invalid.py

Module: Login

Priority: High

Last Execution: Failed

Execution Time: 1 min

3. Script Name: test_payment_retry.py

Module: Payment

Priority: Critical

Last Execution: Failed

Execution Time: 4 mins

4. Script Name: test_cart_add_item.py

Module: Cart

Priority: Medium

Last Execution: Passed

Execution Time: 2 mins

5. Script Name: test_session_timeout.py

Module: Session Management

Priority: High

Last Execution: Failed

Execution Time: 3 mins

6. Script Name: test_profile_update.py

Module: User Profile

Priority: Low

Last Execution: Passed

Execution Time: 1 min

Flaky Scripts:

- test_payment_retry.py

Frequently Failed Modules:

- Login

- Payment

2) Added the PROMPT TEMPLATE in LangFlow and also pasted the prompt which is used in the flow

You are a QA Regression Optimization Expert and Analyze the release changes and optimize the regression suite.

Tasks:

1. Identify impacted modules
2. Select relevant regression scripts for execution
3. Exclude unrelated regression scripts
4. Recommend/Suggest additional testscripts
5. Perform Risk Analysis and Prioritization
6. Generate Regression Coverage Summary and Estimation

Release Notes:

{release_notes}

Change Logs:

{change_logs}

Regression Suite Metadata:

{regression_suite_metadata}

Rules:

- If authentication/payment modules change → Risk = Critical
- If UI-only changes occur → Risk = Medium
- If no business logic changes → Risk = Low

Analyze the release changes and provide:

1. Impacted modules
2. Risk level
3. Recommended regression scope

{release_notes}

Return response in this format:

SECTION 1:

STRICT JSON ONLY

SECTION 2:

Generate a PROFESSIONAL markdown report.

Formatting Rules:

- Use clear section headings with #
- Use tables for script lists
- Use bullet points for recommendations
- Add emojis for risk indicators
- Keep spacing clean
- Use horizontal separators ---
- Highlight important values in bold
- Do NOT return plain paragraphs

Output Format:

Regression Optimization Report

Release Summary

- Item 1
- Item 2

Risk Level

🔴 Critical

Selected Regression Scripts

Script Name	Module	Reason
-----	-----	-----
script	module	reason

Excluded Scripts

| Script Name | Reason |

|-----|-----|

Suggested New Scripts

| Script | Purpose |

|-----|-----|

Coverage Summary

| Metric | Value |

|-----|-----|

Recommendations

- Recommendation 1
- Recommendation 2

3) Added the LLM as Groq and Model as “openai/gpt-oss-120b”

4) Added “Text Output” ; Run them and got the below results

****SECTION 1:****

```json

{

  "impacted\_modules": [

    "Login",

    "Authentication",

    "Payment",

    "Cart",

    "Checkout",

    "Session Management"

],

  "risk\_level": "Critical",

  "selected\_scripts": [

    {

      "script\_name": "test\_login\_valid.py",

      "module": "Login",

      "reason": "Validates successful login flow including new MFA OTP handling"

    },

    {

      "script\_name": "test\_login\_invalid.py",

      "module": "Login",

      "reason": "Covers invalid credentials, OTP failures and resend-OTP functionality"

    },

    {

      "script\_name": "test\_payment\_retry.py",

      "module": "Payment",

```

 "reason": "Tests enhanced retry logic, limit validation and timeout handling (flaky, needs
rerun)"
 },
 {
 "script_name": "test_cart_add_item.py",
 "module": "Cart",
 "reason": "Ensures item addition works with the redesigned responsive UI"
 },
 {
 "script_name": "test_session_timeout.py",
 "module": "Session Management",
 "reason": "Validates extended session timeout and automatic inactivity logout"
 }
],
"excluded_scripts": [
 {
 "script_name": "test_profile_update.py",
 "reason": "Unrelated to changed modules"
 }
],
"suggested_scripts": [
 {
 "script_name": "test_login_mfa_flow.py",
 "purpose": "End-to-end MFA OTP generation, validation and session creation"
 },
 {
 "script_name": "test_login_resend_otp.py",
 "purpose": "Verify resend-OTP limits and throttling behavior"
 },
 {
 "script_name": "test_cart_quantity_update.py",
 "purpose": "Validate quantity increment/decrement logic on responsive UI"
 },
 {
 "script_name": "test_cart_responsive_layout.py",
 "purpose": "Check cart page layout across mobile, tablet and desktop viewports"
 },
 {
 "script_name": "test_payment_retry_limit.py",
 "purpose": "Assert retry-limit enforcement and proper error messaging"
 },
 {
 "script_name": "test_checkout_api_response.py",
 "purpose": "Confirm improved API handling and error scenarios in checkout flow"
 },
 {
 "script_name": "test_session_inactivity_logout.py",
 "purpose": "Ensure automatic logout after 30 min of inactivity"
 }
]

```

```

 }
],
 "coverage_summary": {
 "total_scripts_in_suite": 120,
 "selected_scripts_count": 5,
 "excluded_scripts_count": 1,
 "suggested_scripts_count": 7,
 "estimated_execution_time_minutes": 12,
 "estimated_additional_time_minutes": 21,
 "total_estimated_time_minutes": 33,
 "critical_coverage_percent": 4.17,
 "overall_coverage_percent": 10.0
 },
 "risk_analysis": {
 "authentication": "Critical – new MFA flow introduces security-critical changes",
 "payment": "Critical – modified retry logic and limit validation affect transaction integrity",
 "cart_ui": "Medium – UI redesign and responsiveness changes",
 "session_management": "Medium – timeout increase and inactivity logout"
 }
}
...

```

## **\*\*SECTION 2:\*\***

### **# Regression Optimization Report**

#### **## Release Summary**

- **\*\*Version:\*\*** v6.2.0 (10-May-2026)
- **\*\*New Features:\*\*** MFA for login, enhanced payment retry, responsive cart UI
- **\*\*Enhancements:\*\*** Session timeout extended to 30 min, checkout API response handling improved
- **\*\*Bug Fixes:\*\*** Duplicate payment, cart quantity update, intermittent logout

---

#### **## Risk Level**

 **\*\*Critical\*\*** – authentication and payment modules have business-logic changes; any regression here can impact security and revenue.

---

#### **## Selected Regression Scripts**

| Script Name           | Module | Reason                                                         |
|-----------------------|--------|----------------------------------------------------------------|
| `test_login_valid.py` | Login  | Validates successful login flow including new MFA OTP handling |

|                           |                    |                                                                                        |
|---------------------------|--------------------|----------------------------------------------------------------------------------------|
| `test_login_invalid.py`   | Login              | Covers invalid credentials, OTP failures and resend-OTP functionality                  |
| `test_payment_retry.py`   | Payment            | Tests enhanced retry logic, limit validation and timeout handling (flaky, needs rerun) |
| `test_cart_add_item.py`   | Cart               | Ensures item addition works with the redesigned responsive UI                          |
| `test_session_timeout.py` | Session Management | Validates extended session timeout and automatic inactivity logout                     |

---

## ## Excluded Scripts

|                          |                              |
|--------------------------|------------------------------|
| Script Name              | Reason                       |
| ----- -----              |                              |
| `test_profile_update.py` | Unrelated to changed modules |

---

## ## Suggested New Scripts

|                                     |                                                                    |
|-------------------------------------|--------------------------------------------------------------------|
| Script                              | Purpose                                                            |
| ----- -----                         |                                                                    |
| `test_login_mfa_flow.py`            | End-to-end MFA OTP generation, validation and session creation     |
|                                     |                                                                    |
| `test_login_resend_otp.py`          | Verify resend-OTP limits and throttling behavior                   |
| `test_cart_quantity_update.py`      | Validate quantity increment/decrement logic on responsive UI       |
|                                     |                                                                    |
| `test_cart_responsive_layout.py`    | Check cart page layout across mobile, tablet and desktop viewports |
|                                     |                                                                    |
| `test_payment_retry_limit.py`       | Assert retry-limit enforcement and proper error messaging          |
| `test_checkout_api_response.py`     | Confirm improved API handling and error scenarios in checkout flow |
|                                     |                                                                    |
| `test_session_inactivity_logout.py` | Ensure automatic logout after 30 min of inactivity                 |

---

## ## Coverage Summary

|                                             |            |
|---------------------------------------------|------------|
| Metric                                      | Value      |
| ----- -----                                 |            |
| **Total scripts in suite**                  | 120        |
| **Selected scripts**                        | 5          |
| **Excluded scripts**                        | 1          |
| **Suggested new scripts**                   | 7          |
| **Estimated execution time (selected)**     | 12 min     |
| **Estimated additional time (new scripts)** | 21 min     |
| **Total estimated regression time**         | **33 min** |

| **Critical-module coverage** | 4.17 % |  
| **Overall coverage of impacted area** | 10.0 % |

---

## ## Recommendations

- **Run the selected 5 scripts plus all 7 suggested new scripts** to achieve ~10 % coverage of the impacted area, focusing on critical authentication and payment paths.
- **Stabilize flaky script `test\_payment\_retry.py`** (e.g., add retries, improve wait strategies) before the regression cycle.
- **Prioritize execution order:** MFA flow → Payment retry → Session timeout → Cart UI checks.
- **Allocate ~35 minutes** for the full regression run (includi...